HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES
Actuarial Mathematics and Statistics

F78PA
Probability and Statistics A

Semester 1 - 2018/19
Duration: 2 hours

Question	Marks
1	21
2	16
3	12
4	17
5	9
6	10
Total Marks	85

Candidates must attempt ALL SIX questions.
To receive full credit you must show your work and explain your answers.

Approved electronic calculators may be used

AMS Formula Sheet and Cambridge Statistical Tables are provided

RJRC F78PA

1. Consider the function

$$F_X(x) = \begin{cases} \frac{1}{3x^4} & \text{if } x \leq -1 \\ \frac{1}{3} & \text{if } -1 < x < 1 \\ 1 - \frac{c}{x^2} & \text{if } 1 \leq x \end{cases}$$

(a) State the 4 properties of a distribution function. [2 marks]

(b) Find all possible values for c such that the function F_X is a distribution function for a random variable X. Give reasons for your answer by referring to the properties of distribution functions. [3 marks]

For the following questions fix $c = 2/3$ so that

$$F_X(x) = \begin{cases} \frac{1}{3x^4} & \text{if } x \leq -1, \\ \frac{1}{3} & \text{if } -1 < x < 1, \\ 1 - \frac{2}{3x^2} & \text{if } 1 \leq x. \end{cases}$$

and let F_X be the distribution function for X.

(c) Sketch F_X in this case and explain why X has a density function. Show the density function for X is given by the following function:

$$f_X(x) = \begin{cases} \frac{-4}{3x^5} & \text{if } x < -1, \\ 0 & \text{if } -1 < x < 1, \\ \frac{4}{3x^3} & \text{if } 1 \leq x. \end{cases}$$

[4 marks]

(d) Explain why the image of the random variable X (ImX) is $(-\infty, -1] \cup [1, \infty)$. [1 marks]

(e) Find $E(X)$, the expected value of X. [3 marks]

(f) Suppose that $Y = X^2$.

i) Explain why the image of the random variable Y (ImY) is $[1, \infty)$ and show that for $1 \leq y$

$$F_Y(y) = F_X(\sqrt{y}) - F_X(-\sqrt{y})$$

where F_Y is the cumulative distribution function of Y. [4 marks]

ii) Use part (f)(i) to find the density for Y. [4 marks]

[Total 21 marks]

[PLEASE TURN OVER]

2. Let X and Y be two random variables with joint probability density function

$$f_{X,Y}(x,y) = \begin{cases} \frac{3}{4}y, & \text{for } 0 \leq x, y \text{ such that } 0 \leq x+y \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Sketch the region where the density $f_{X,Y}(x,y)$ is non-zero.

[1 mark]

(b) Find the marginal density function $f_X(x)$ for X and show that the conditional density of Y given X is given by

$$f_{Y|X}(y|x) = \frac{2y}{(2-x)^2} \quad \text{for } 0 \leq x, y \text{ such that } 0 \leq x+y \leq 2.$$

[4 marks]

Compute:

i) $P(Y < 1 \mid X = 1/2)$ [2 marks]

ii) $E(Y \mid X = 1/2)$. [2 marks]

(c) Let $Z = X + Y$. State the image of Z and find the distribution function $F_Z(z)$ and density function $f_Z(z)$ of Z. [7 marks]

[Total 16 marks]

[PLEASE TURN OVER]

3. Bill drives to work each day and and his journey time is X minutes, where $X = 15 + Y$ and Y is exponentially distributed with mean 10 (i.e., $Y \sim$Exp(1/10)).

(a) One day Bill leaves home at 8:45am for a meeting with Stan. Stan will arrive at the meeting at W minutes after 9am where W is uniformly distributed on the interval $(0, 15)$ (i.e. $W \sim U(0, 15)$). The start time of the meeting will be the time, T, in minutes after 9am when both Stan and Bill have arrived.

i) Explain why $T = \max(W, Y)$ and why the image of T, $ImT = [0, \infty)$. [2 marks]

ii) Assume W and Y are independent.

Find F_T, the distribution function of T . [4 marks]

Hence or otherwise, find $P(T < 15|T < 30)$, the probability that the meeting starts before 9:15am given it starts before 9:30am. [3 marks]

(b) Another day Bill has a meeting with his boss at 9am. What time does Bill need to leave in the morning to guarantee that he is late with a the probability less than 0.01? [3 marks]

[Total 12 marks]

[PLEASE TURN OVER]

4. Suppose that a random variable X has a $Gamma(\alpha, \lambda)$ distribution.

(a) Show the moment generating function for X is

$$M_X(\theta) = \left(\frac{\lambda}{\lambda - \theta}\right)^{\alpha},$$

for $\theta < \lambda$. [4 marks]

(b) Suppose that $Y = cX$, where c is a positive constant. Show that $M_Y(\theta) = M_X(c\theta)$, where M_X and M_Y are the moment generating functions for X and Y respectively.
Hence, or otherwise, find $M_Y(\theta)$ and determine the distribution of Y. [5 marks]

[Total 9 marks]

5. Each day the number of customers visiting a department store, N, follows a Poisson distribution with mean 150. We assume that each customer visits the home-ware department with probability 0.3, independent of all other customers. Let N_H be the number of customers who visit the home-ware department in a day.

(a) By writing N_H as a compound sum, i.e. $N_H = \sum_{i=1}^{N} X_i$ where X_i are independent Bernoulli(0.3) random variables, and using moment generating functions, show that N_H has Poisson distribution with mean 45. [5 marks]

(b) Each day the home-ware department is given a performance score,S, between 0 and 1. From historic data, it is known that the performance score is dependent on the number of customers visiting the home-ware department. Given the department receives n customers in a day, i.e given $N_H = n$, the conditional distribution of the performance score, S, is

$$F_{S|N_H=n}(s|n) = \begin{cases} 0 & \text{for } s < 0, \\ s^{n/10} & \text{for } 0 \leq s < 1, \\ 1 & \text{for } 1 \leq s. \end{cases}$$

Show that distribution function of S is given by

$$F_S(s) = \begin{cases} 0 & \text{for } s < 0, \\ e^{45(s^{1/10}-1)} & \text{for } 0 \leq s < 1, \\ 1 & \text{for } 1 \leq s, \end{cases}$$

and sketch F_S. [5 marks]

[Total 10 marks]

[PLEASE TURN OVER]

6. **(a)** Let X and Y be independent random variables with densities f_X and f_Y respectively. Show that $E(XY) = E(X)E(Y)$. [3 marks]

Let $X_1, X_2, \ldots, X_{200}$ be independent identically distributed random variables with density function:

$$f_X(x) = \begin{cases} \frac{1}{2x} & \text{if } \frac{1}{e} \leq x \leq e \\ 0 & \text{otherwise} \end{cases}$$

(b) Find the expected value of X_1X_2, $E(X_1X_2)$. [3 marks]

(c) Let $Y_i = \ln(X_i)$. Explain why $ImY = [-1, 1]$ and show that Y_i is uniformly distributed on the interval $[-1, 1]$. [5 marks]

(d) State the Central Limit Theorem and use it to estimate the probability

$$P\left(\prod_{i=1}^{200} X_i < 0.083\right).$$

[6 marks]

[**Hint**: Recall that $\ln(\prod_{i=1}^{200} X_i) = \sum_{i=1}^{200} \ln(X_i)$]

[Total 17 marks]

[END OF PAPER]